Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

Naim Rashid, PhD

Department of Biostatistics · Lineberger Comprehensive Cancer Center · UNC

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Today’s argument

> **New RAS therapies in PDAC need new kinds of trials.**

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

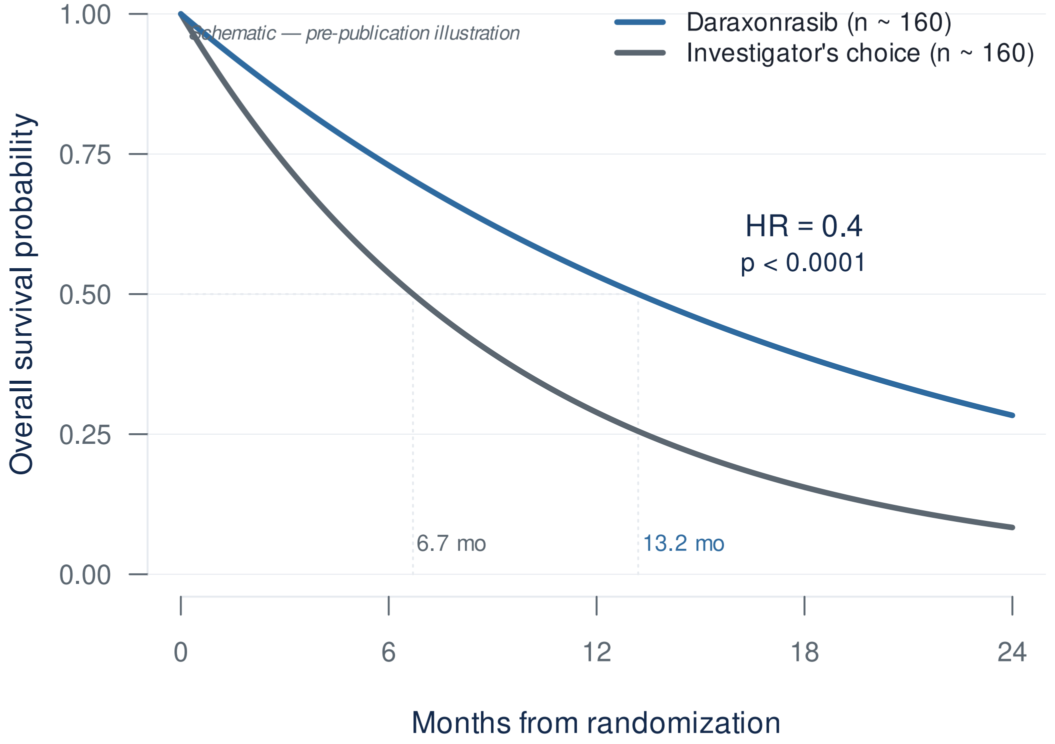
> **The difficulty: getting them calibrated and launched.**

The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

- **RASolute 302** (2L mPDAC, *NEJM* 2026)[1]:
  - **Median OS: 13.2 vs 6.7 months**
  - **HR 0.40 — 60% reduction in risk of death** (p < 0.0001)
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



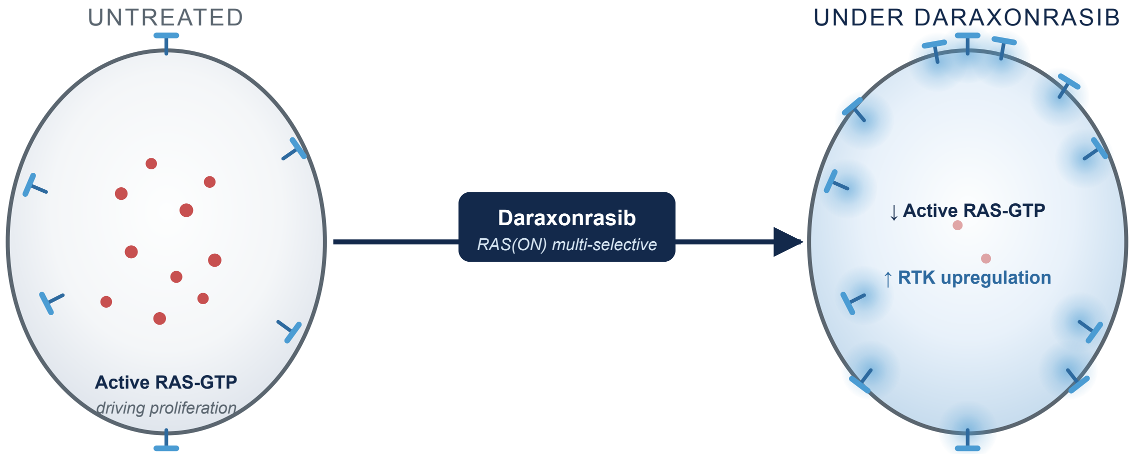
…but monotherapy has visible ceilings

RMC-6236-001 (PDAC cohort, NEJM 2026)[2]

> **Half of patients on monotherapy at 300 mg already need dose adjustments.** Combinations layer on top — with less room to spare.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations[3]
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40)[4]:
  - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



RTK upregulation is simultaneously a resistance mechanism and a rational combination target

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated[5,6].
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics[7], it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the "joint surface") is never characterized.

Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair — not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

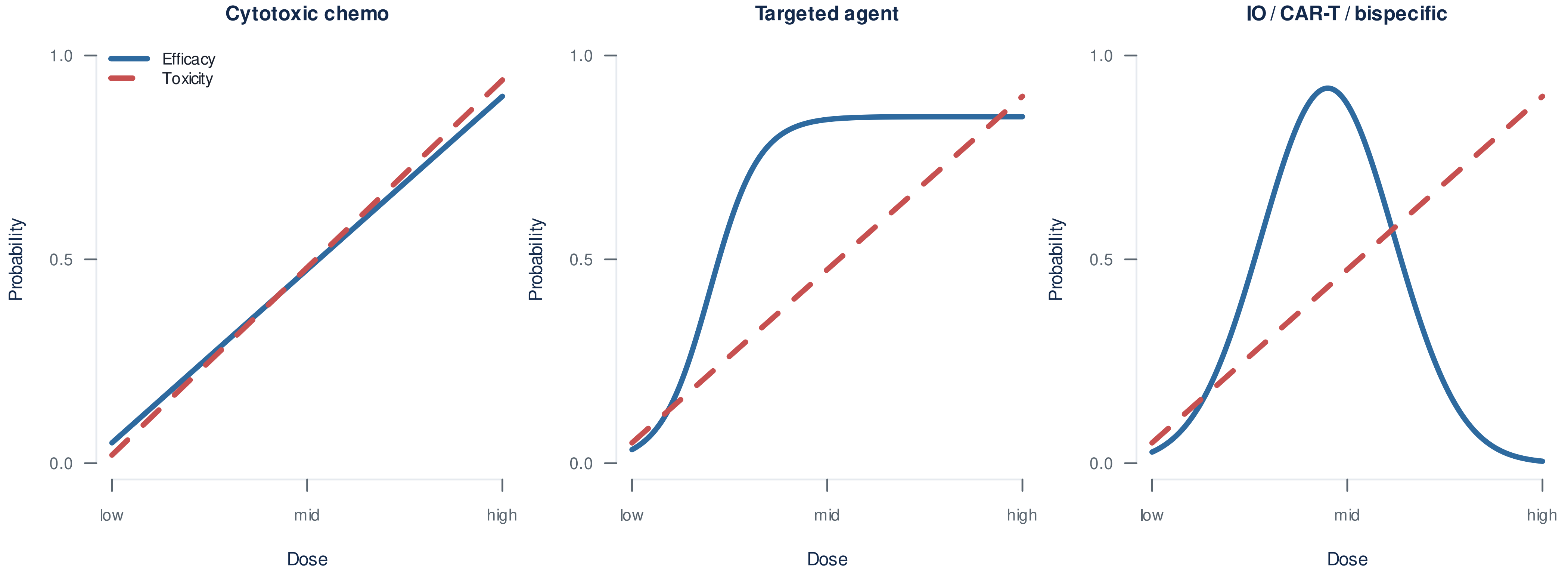
- **Monotherapy (CodeBreaK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window[8]

> **What MTD-only dose-finding was never built to handle, all at once:** joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)[10]:
  - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
  - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure[11]
- **MTD ≠ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions[12]

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

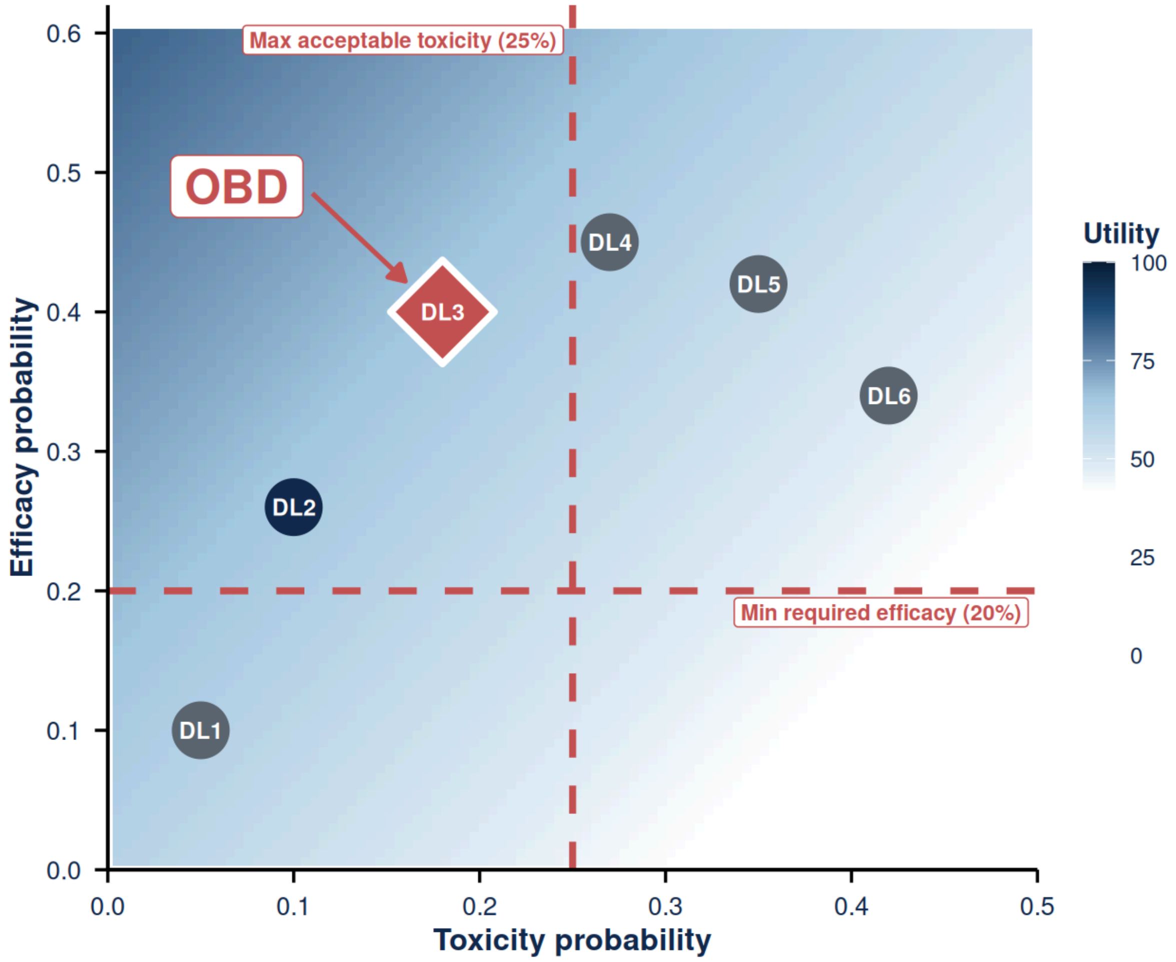
Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

"A response is worth this; a Grade 3 toxicity costs this."

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)[14]:

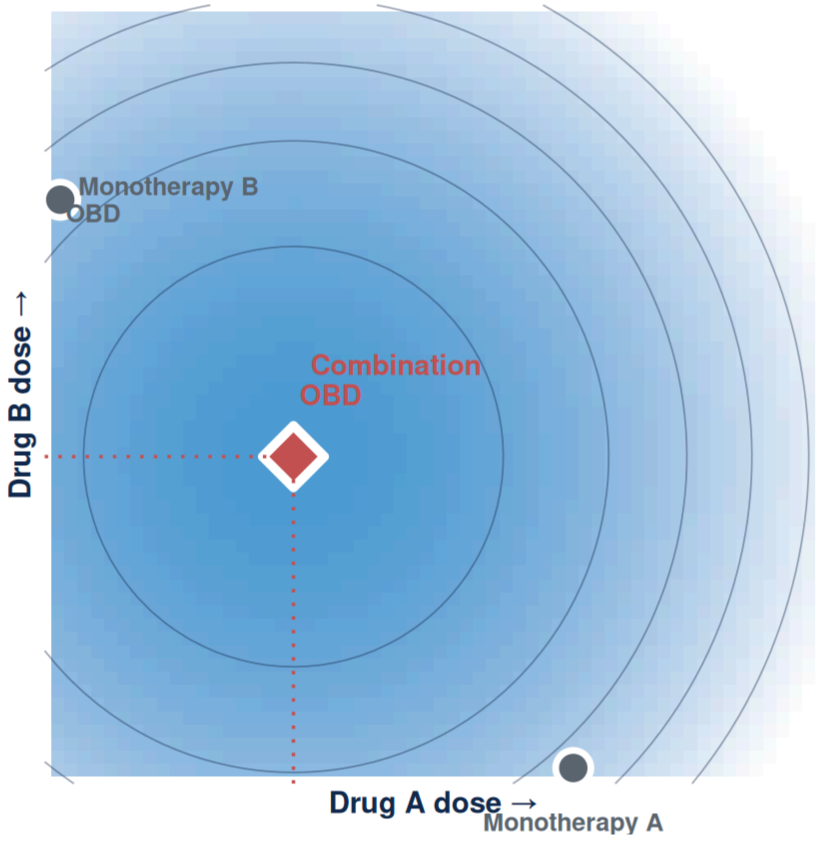
	TOX	NO TOX
Response	40	**100**
No response	**0**	60



OBD = admissible dose (inside red lines) with the highest utility.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

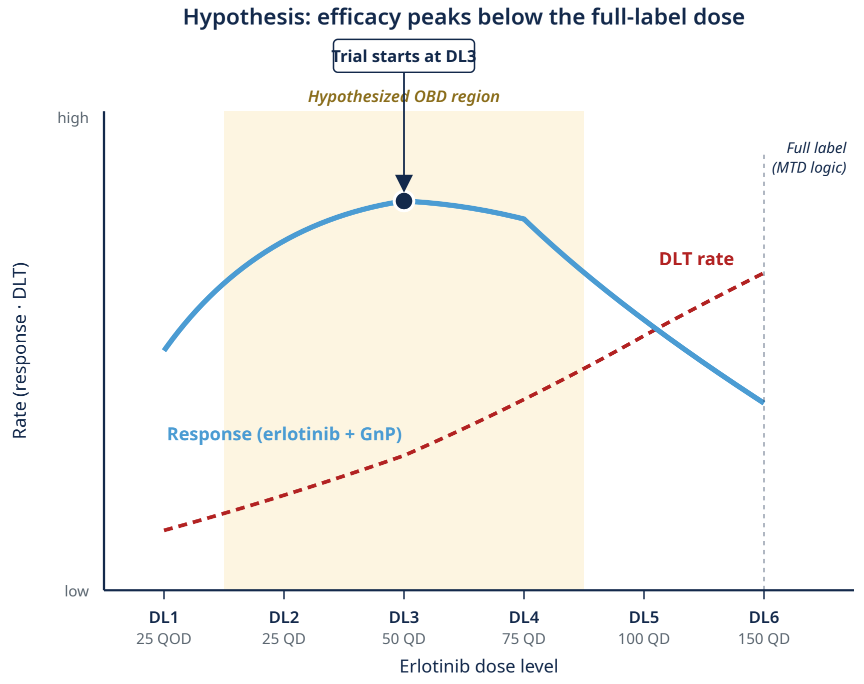
> **Fix what you know. Search what you don't.**

RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface[4].

Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off[14,17].
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

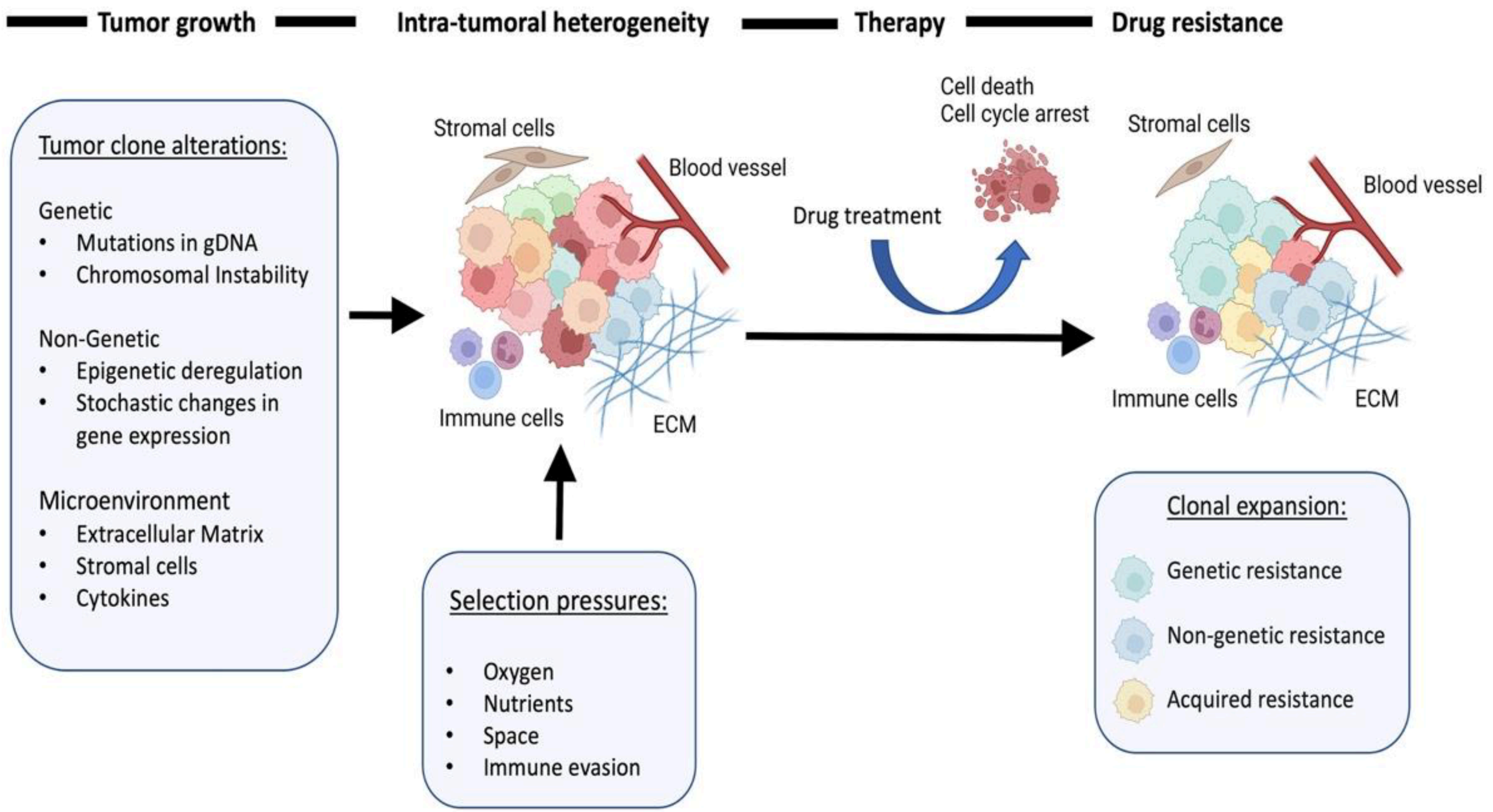
Trial: LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurIST[18] · combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data[19] · **up to 52 patients; ~28 expected at the OBD.**



Resistance is the next problem

Phase 2 onward: *each patient is a moving target*

Resistance is a dynamic process

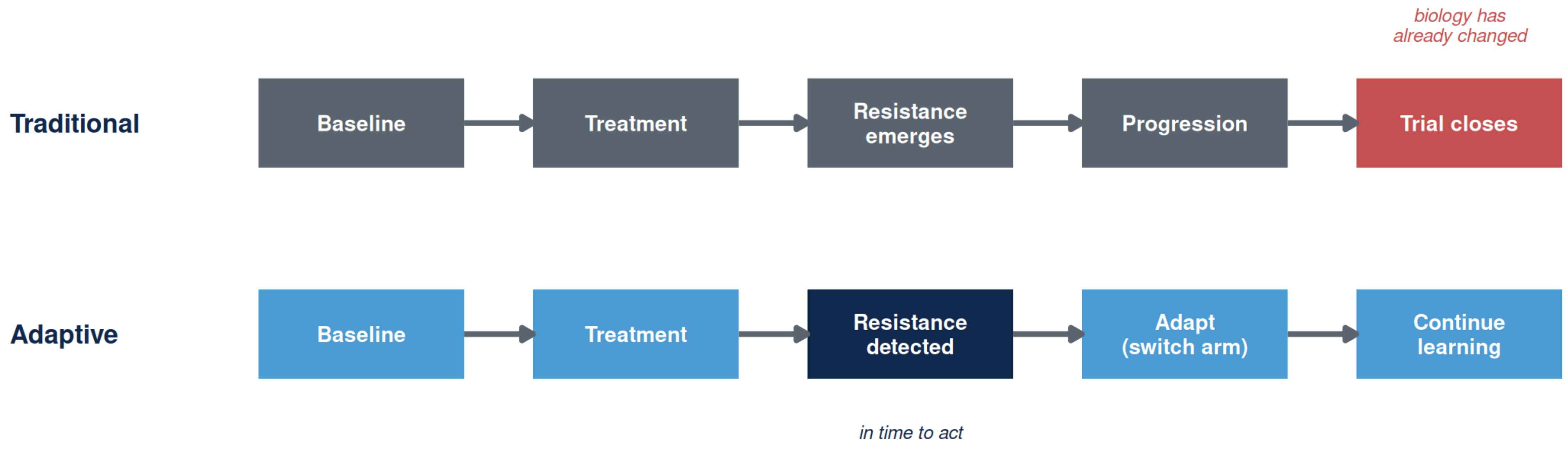


Adapted from Karagiannis & Rampias, Cancers 2022[20].

Resistance mechanisms emerge under RAS-inhibitor pressure

> **Daraxonrasib monotherapy in mPDAC** (n=44, paired ctDNA): **59% show RAS-pathway resistance at progression — KRAS amplification in 36%**[21]

Why classical trial designs miss resistance



RESISTANCE TYPE	WHAT YOU NEED
Genomic (KRAS amplification, NRAS/BRAF/MAP2K1)	ctDNA
Cell-state / plasticity (basal↔classical, partial EMT)	biopsy + transcriptomics + imaging

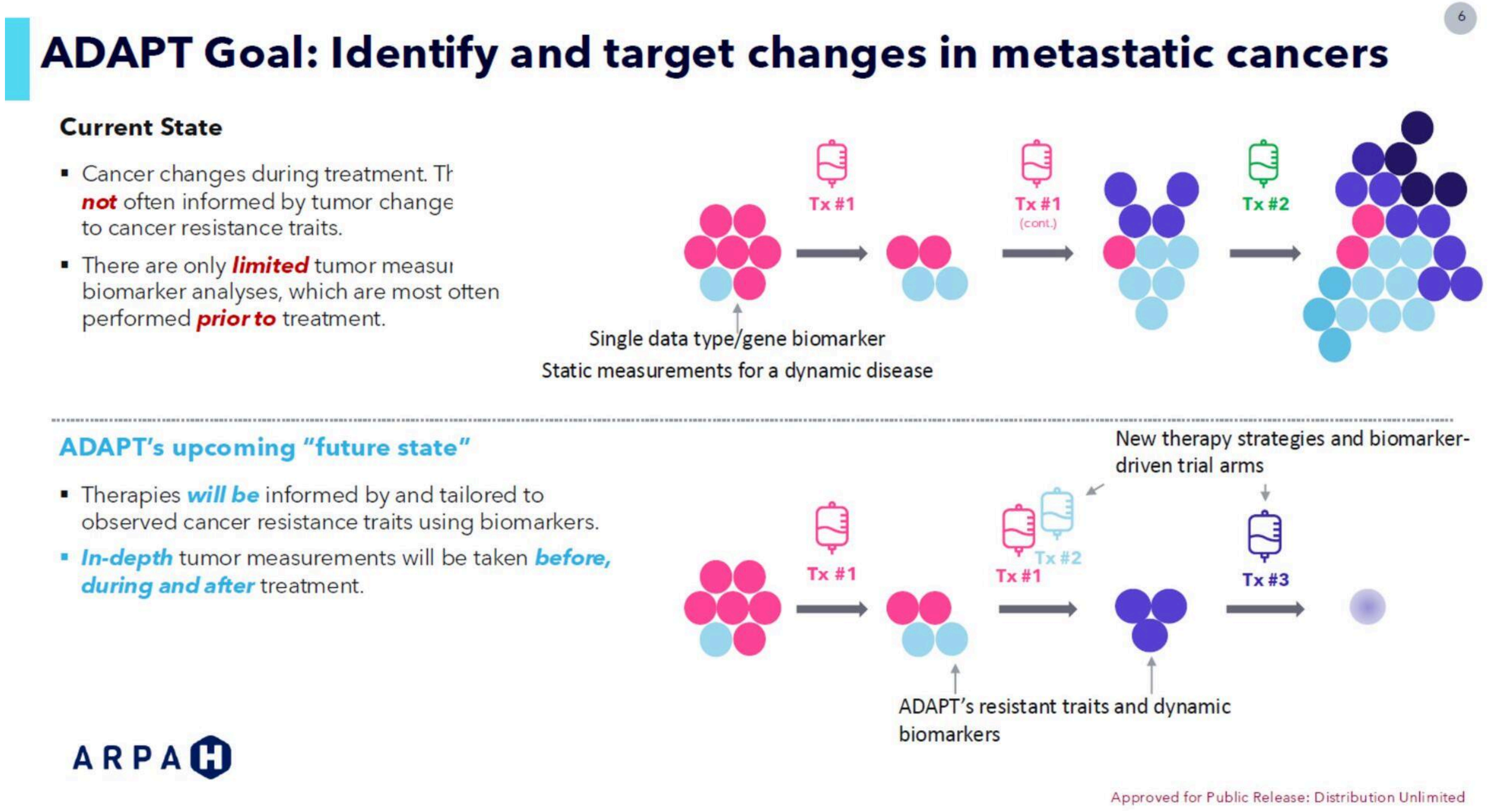
A resistance-aware trial has to **do both** during enrollment[22,23].

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

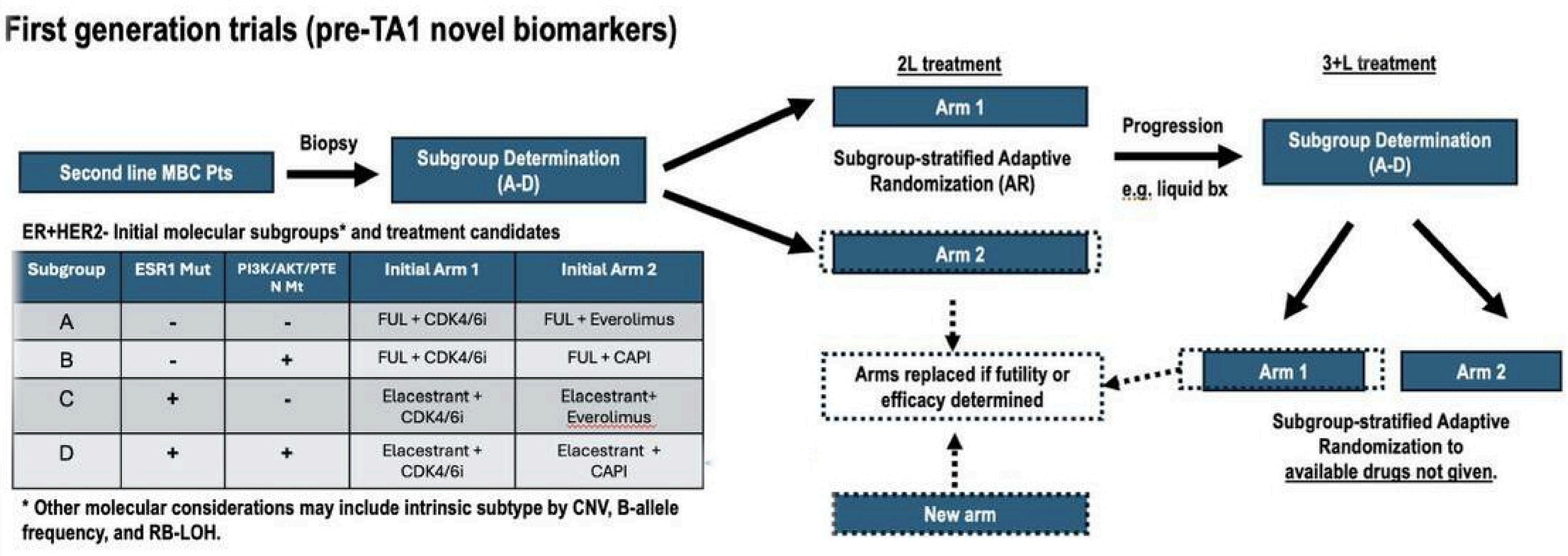
*ARPA-H **ADAPT** → **EVOLVE** (UNC-led; Carey MPI, Rashid MPI). **Enrolling.***



ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC

EVOLVE is the operational template; PDAC is the next application.



Subgroup	ESR1 Mut	PI3K/AKT/PTEN Mt	Initial Arm 1	Initial Arm 2
A	-	-	FUL + CDK4/6i	FUL + Everolimus
B	-	+	FUL + CDK4/6i	FUL + CAPI
C	+	-	Elacestrant + CDK4/6i	Elacestrant+ Everolimus
D	+	+	Elacestrant + CDK4/6i	Elacestrant + CAPI

* Other molecular considerations may include intrinsic subtype by CNV, B-allele frequency, and RB-LOH.

TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

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EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurIST[18], same Bayesian arm-swap logic on daraxonrasib-combination arms.

> **Same architecture, different disease.**

What this demands of the statistical design

Four design axes any such trial needs.

Adaptive stopping	Real-time biomarker discovery	Longitudinal monitoring	In-trial evaluation
Bayesian futility / efficacy boundaries at each interim	Pre-specified signatures + agnostic discovery window	Serial ctDNA, imaging, biopsies on every patient	Biomarker-guided cohorts within the same protocol

Methods are ready. We're operationalizing them in PDAC trials — starting with PANGEA.

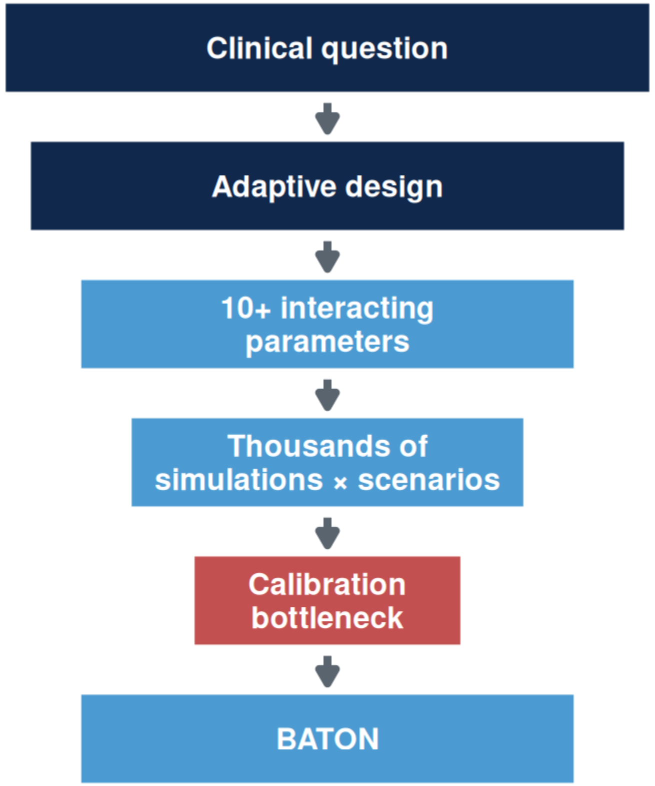
Putting these into practice

Why don’t we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

*Calibration tunes a design to hit its **operating characteristics** — the trial’s report card: false-positive control, power, expected enrollment, early stopping, and time to readout.*



Methods adopted. Dose-optimization plans — where calibration cost bites — not[24].

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving all four simultaneously.

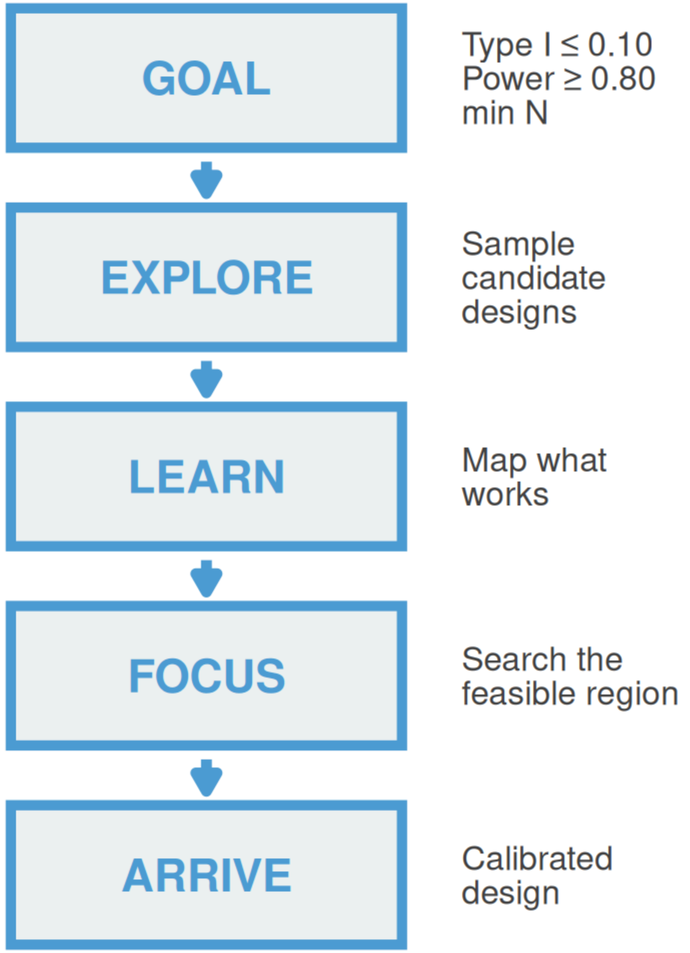
1. FDA compliance Power ≥ 0.80 Type I ≤ 0.10	**2. Minimize patients** Smaller E[N] Efficient use of scarce patients
3. Stop bad early Futility stopping Protect patients from inactive arms	**4. Accelerate good** Efficacy stopping Faster access when drug works

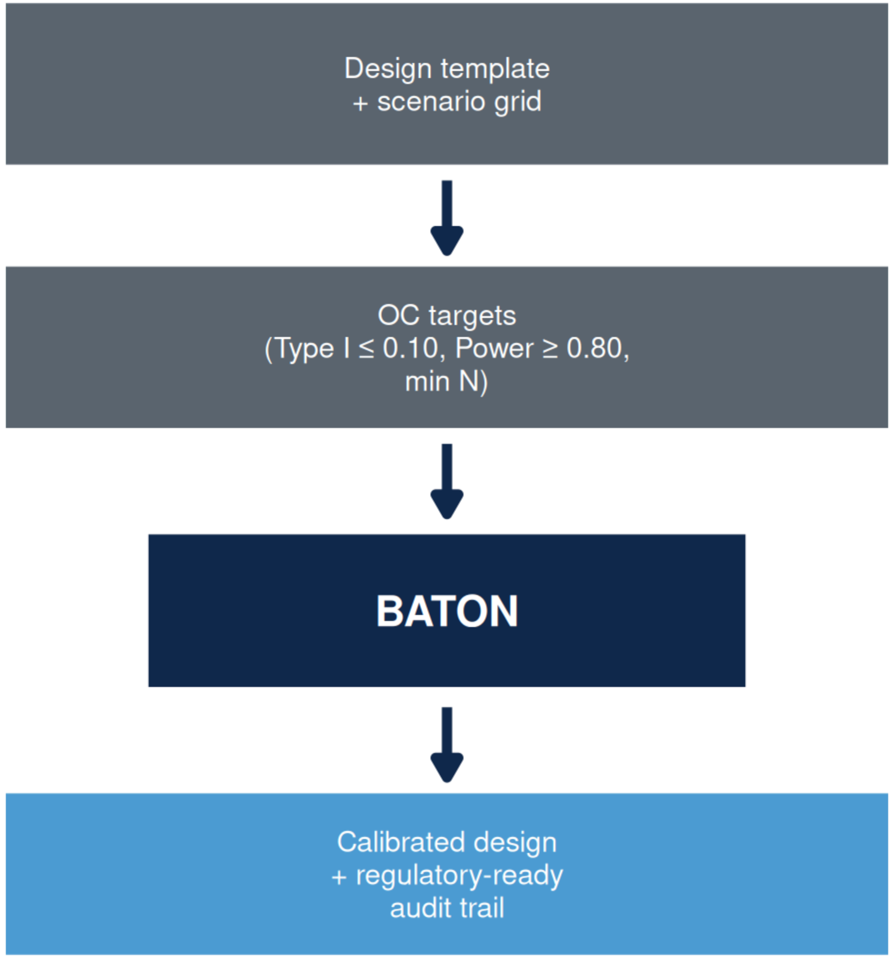
"We could design the science. We needed a faster way to calibrate the trial."

BATON: the calibration tool

BATON = **B**ayesian **A**daptive **T**rial **O**ptimizatio**N**[25]

BATON turns clinically necessary trial complexity into calibrated, launchable designs.





You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates by simulation and returns the ones that meet your targets. R package: github.com/naimurashid/BATON[25]

A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓ Type I 7.0% (target ≤10%) Power 82.2% (target ≥80%)	**2. Minimize patients ✓** ~23 patients spared (avg N 53 vs Max N 76)
3. Stop bad early ✓ 78% early futility stop when drug is inactive	**4. Accelerate good ✓** 81% early efficacy stop when drug works

70% fewer evaluations than grid search — same optimum. Manual calibration: weeks to months. BATON: hours, with a regulatory-ready audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurIST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT:** GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)[26]
- **BATON-calibrated** four parameters under Type I ≤ 0.10, Power ≥ 0.80[25]

Achieved operating characteristics:

METRIC	VALUE
Power	**82.2%**
Type I error	7.0%
Max N	**76** (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop | H_0)	**77.9%**
Pr(early efficacy stop | H_1)	**81.4%**
Expected duration	27.3 mo

H_0 = drug inactive (no PFS benefit over GnP alone) · H_1 = drug active at the assumed effect size (HR 0.58)

Read as: 78% chance of stopping early when the drug is inactive; 81% chance of stopping early when it works.

Without BATON: four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

An unexpected finding: *caps set too low suppress early-efficacy declarations — even when the drug works, the trial waits for the final analysis to say so. BATON returned the* ***best trade-off curve*** *across candidate caps; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.*

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:

PDAC

RAS-backbone combinations OBD selection

CRC

Joint-surface novel-novel doublets

Basket / umbrella

Biomarker-stratified subtrials shared decision rules

Robust calibration

Designs robust to uncertain trial assumptions

BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON makes them practical to calibrate and launch.**

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Questions